Character Network Extraction from Narrative Texts: An Iterative Approach to Named Entity Recognition and Relation Typing

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Abstract

We present a modular Python pipeline for extracting typed character interaction networks from literary texts written or translated in french. We developed this system iteratively: Starting from Named Entity Recognition (NER) that went from a slow multi-model ensemble (spaCy, Stanza and Flair) to a fast rule-based approach completing all 37 chapters in almost one second; moving to character relation typing that went from NLI-based classifiers (6–8 hours on CPU in Google Colab) through LLM APIs (quota exhausted) to a sentiment-based approach using a multilingual XLM-roBERTa-base model, which processes the full corpus in about 8 minutes with no annotated data. Applied to two French Asimov novels (*Prélude à Fondation* and *Les Cavernes d'Acier*), the pipeline achieves 100% chapter coverage and produces one typed GraphML file for each chapter for a Kaggle competition dedicated to this AMS project. The relations label distribution across 37 chapters is dominated by the hostile label (hostile = 239, neutral = 82, friendly = 58), which is consistent with the political intensity and the constant conflict in Asimov's writings. We document each approach tried, the reason for its failures (if applicable), and the algorithmic errors that remain.

Index Terms: character networks, named entity recognition, sentiment analysis, iterative system design, literary NLP, relation typing

Code: `https://github.com/WaelMansoura/Reseaux-de-personnage`

1. Introduction

Character networks model a narrative as a graph whose nodes are characters and edges capture their interactions. They support computational literary analysis: centrality reveals protagonists, community detection uncovers factions, and temporal comparison tracks alliance shifts across chapters [1, 2].

Early extraction systems produced untyped co-occurrence networks [3]. Typed relation extraction traditionally requires annotated corpora [4], which do not exist for French literary fiction. Recent work has explored zero-shot classification via Natural Language Inference (NLI) [5, 6], but practical deployments face severe runtime constraints on consumer-grade hardware.

This paper makes four contributions. First, a complete modular pipeline from raw French text to typed GraphML character networks. Second, a documented iterative methodology comparing multi-model NER versus rule-based NER and NLI versus sentiment-based relation typing, with runtime and quality trade-offs for each. Third, a systematic error characterisation of every module. Fourth, empirical results on 37 chapters of Asimov fiction showing that hostile interactions dominate the labelled corpus, a finding consistent with the novels' genres.

2. État de l'art

2.1. Co-occurrence networks

The simplest way to link two characters is to check whether they appear close together in the text. Moretti (2005) applied this idea to drama, where shared stage presence makes interaction explicit [1]. Elson et al. (2010) brought it to novels, using dialogue exchanges as the link criterion [3]. Labatut and Bost (2019) surveyed the field and found that co-occurrence remains the dominant approach because it needs no labelled data [7]. The main design choice is how wide to set the window: Grayson et al. (2016) found that 20–300 words covers most interactions in 19th-century prose [10]. The known weakness is over-linking: if two characters share a scene without speaking to each other, a wide window still draws an edge between them.

2.2. Character detection in novels

Standard NER tools are trained on news text, so they struggle with literary fiction. Novels introduce invented names (*Trantor*, *Terminus*), refer to characters by title (*l'Empereur*, *The Director*), and use long chains of pronouns and nicknames for the same person. Vala et al. (2015) documented these problems across many tools and showed that verb-argument patterns can recover character mentions that name-matching alone misses [11]. Dekker et al. (2019) ran a systematic comparison of four tools on 40 novels: the best result was F1 = 0.886 on *Harry Potter*; the worst was 0.348 on *Gardens of the Moon* [8]. Crucially, errors on place names (tagged as persons) survived majority voting because every model made the same mistake, having trained on similar data. That result motivated our switch to a rule-based approach.

2.3. Relation typing

Three families of approaches exist for labelling an edge as friendly, hostile, or neutral.

Supervised classifiers [4] give the best accuracy but require hand-labelled training examples. No such dataset exists for French literary fiction, so this path was closed to us from the start.

Zero-shot NLI asks a language model whether a text snippet entails a given description (e.g., "these two characters are enemies") without any task-specific training [5, 6]. It works, but it is slow: our tests ran for 6–8 hours on CPU for both mDeBERTa-v3 and distilcamembert-base-nli, which made rerunning experiments impractical.

Using sentiment as a proxy is the oldest approach, going back to Shakespeare network studies [12]. The idea is simple: if the text around two characters reads as positive, the relation is friendly; if negative, hostile. The drawback is that literary prose

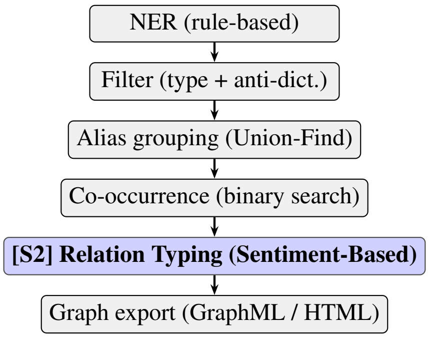

Figure 1: *Final pipeline. Shaded: Semester 2 contribution.*

does not work like social-media text: irony, indirect speech, and polite disagreement all produce weak or misleading sentiment scores [7]. LLMs handle nuance better, but free-tier quotas ran out before we could process all 37 chapters [13].

2.4. French literary NLP

For French, CamemBERT [14] is the reference model, beating multilingual alternatives (XLM-R, mBERT) by over 10 points on NER and NLI in supervised settings. The BookNLP-fr project is building a dedicated French literary pipeline using the *Democrat* corpus of 19th-century novels [15]. Long-LitBank-fr provides coreference annotations for three full-length French novels (285,176 tokens) [16]. Neither dataset includes relation-type labels. No annotated corpus exists for friendly/hostile/neutral character relations in French fiction, which is why we could not train or calibrate a supervised classifier.

2.5. Corpus and Task

The task, framed as a Kaggle competition, requires producing one GraphML file per chapter. Each encodes an undirected character network with node attribute `names` (semicolon-separated surface forms) and edge attributes `weight` (co-occurrence count) and `edge_type` $\in$ {*friendly*, *hostile*, *neutral*}.

The corpus consists of two French translations of Isaac Asimov novels: *Prélude à Fondation* (PAF, 19 chapters `paf0`–`paf18`) and *Les Cavernes d'Acier* (LCA, 18 chapters `lca0`–`lca17`). Both novels have specific NLP challenges: invented proper nouns (*Trantor*, *Terminus*) that NER models tag as persons, and robot names, following the pattern *R. Daneel Olivaw*, and Asimov's expository dialogue style with limited affective vocabulary. LCA follows a detective narrative with explicit antagonist structures; PAF has political intrigue between academic and imperial factions. Together these genre characteristics explain the hostile-dominant label distribution observed in results.

3. Methodology

The pipeline consists of six sequential modules. Figure 1 shows the data flow.

3.1. NER: from ensemble to rule-based

First approach — multi-model ensemble (`nlp_multi_ner.py`). Three models were run independently: `spaCy` (`fr_core_news_lg`), `Stanza` (French model), and optionally `Flair`. A majority vote retained entities tagged as `PER` by $\geq$ 2 models. To boost recall, an `EntityRuler` was injected before the spaCy NER layer, pre-loaded with the full Asimov gazetteer (character and location patterns), ensuring known names were always tagged correctly regardless of context.

Two problems emerged. First, runtime: downloading and initialising three models added significant setup overhead, and full processing of 37 chapters took several minutes, too slow for iterative parameter tuning. Second, false positives survived the vote: both spaCy and Stanza consistently tagged fictional place names (*Terminus*, *Trantor*) and institutional nouns (*l'Empire*) as `PER` and even some common nouns when capitalised at the start of a chapter or after a colon (*la Crise*, *le Conseil*), so majority vote did not filter them.

Final approach — rule-based NER (`nlp_manual_ner.py`). Four strategies are combined in priority order:

1. *Gazetteer lookup*: a curated dictionary of characters and locations from both novels, applied via longest-match non-overlapping scan with French word-boundary regex.
2. *Capitalisation heuristics*: tokens capitalised mid-sentence that do not appear in lowercase elsewhere in the text, are absent from the anti-dictionary, and are not already matched by the gazetteer. Consecutive capitalised tokens are grouped into multi-word candidate names.
3. *Regex patterns*: titled names (*Docteur X*, *Professeur Y*, *Commissaire Z*) and robot convention (*R. Daneel*, *R. Giskard*).
4. *Dynamic blocklist*: words with capitalisation ratio ≥ 0.70 and total count ≥ 5 across the full corpus that are not in the gazetteer are automatically added to a blocklist. Heuristic-only names must appear $\geq$ 2 times (`MIN_CAP_FREQ`) to be retained.

This variant processes all 37 chapters in $\approx$1 second with no model download, enabling rapid iteration. It is the approach used in the final submission.

3.2. Filtering

A two-stage filter removes false positives. First, a type filter retains only `PER` entities. Second, an anti-dictionary of 1012 entries covers Asimov planet and city names, institutional nouns, and high-frequency false positives. Additional validity rules in `is_valid_entity()` reject ALL-CAPS tokens, strings of 1–2 characters, and hyphen-only forms.

3.3. Alias resolution (`nlp_aliases.py`)

Three layers resolve variant surface forms to canonical identities:

Layer 1 — automatic Union-Find merge. Names are normalised (lowercase, title removal, punctuation stripping, accent preservation) before keyword extraction. The function `_should_merge` applies: (i) *subset rule* for two multi-word names: merge only if one keyword set is a proper subset of the other, preventing "Bayta Darell" from merging with "Arcadia Darell" despite the shared surname; (ii) *single-to-multi rule*: a single token merges into a multi-word name that contains it as a meaningful keyword; (iii) *fuzzy fallback*: for single-word pairs, require `rapidfuzz` ratio $\geq$ 88.

An *ambiguity guard* runs before the main merge pass: if a bare token (e.g., *Darell*) appears as keyword in multiple distinct multi-word names, it is pre-assigned only to the most-frequent one, preventing a shared surname from incorrectly unifying two different characters.

Layer 2 — gazetteer-forced grouping (`apply_gazetteer_aliases`): known alias groups from the Asimov gazetteer are merged unconditionally.

Layer 3 — manual overrides (`apply_manual_aliases`): hand-crafted pairs for cases that fuzzy matching cannot reach (e.g., *Empereur* $\rightarrow$ *Cléon*).

3.4. Co-occurrence detection (`nlp_cooccurrence.py`)

The chapter text is tokenised with `re.findall(r"\w+", text.lower())`; character names are tokenised identically for exact match. Character mention positions are precomputed; `bisect_left` provides $O(\log N)$ window membership testing. Each window of `distance_max` = 150 tokens where both A and B are present increments the edge weight $w(A, B)$. Pairs with weight that are less than `min_occurrences` = 2 are discarded.

3.5. Relation typing: four approaches, one retained

Approach 1 — NLI with mDeBERTa (`mDeBERTa-v3-base-mnli-xnli`, ≈350 MB). Zero-shot classification: three French hypotheses per snippet scored for entailment probability, averaged across up to 5 snippets, with a confidence fallback to *neutral*. **Problem: 6–8 hours** to process both books on Google Colab CPU. Completely impractical for iteration. Rejected.

Approach 2 — NLI with CamemBERT (`cmarkea/distilcamembert-base-nli`, ≈110 MB, French-specific, lighter). Same NLI procedure, lighter model. **Problem: same order-of-magnitude runtime.** Rejected.

Approach 3 — LLM APIs (Gemini, OpenRouter free tier). Batch prompting: classify the relation for each character pair given the extracted snippets. **Problem: free-tier rate limits and token quotas exhausted** before all 37 chapters could be processed. Rejected.

Approach 4 — Sentiment-based classifier (final) (`twitter-xlm-roberta-base-sentiment`) [9]. Multilingual sentiment model (positive / neutral / negative). For each snippet the input is formatted as `"{charA} et {charB}: {snippet[:400]}"`, passed through the tokeniser and model, then softmax is applied to the logits. A scalar sentiment score is computed as $s = p_{\text{pos}} - p_{\text{neg}}$; the mapping is: $s > 0.1 \Rightarrow$ *friendly*; $s < -0.1 \Rightarrow$ *hostile*; otherwise *neutral*. A weighted majority vote is taken across up to `MAX_CONTEXTS_PER_PAIR` = 5 snippets (each weighted by $|s|$); the label with the largest accumulated weight wins. If no snippets are extracted for a pair, the label defaults to *neutral*. **Runtime: ≈470 s (∼8 minutes) for 37 chapters on CPU.**

Results are cached in `relation_cache.pkl`, keyed by `(chapter_id, canon_A, canon_B)` with $\text{canon}_A < \text{canon}_B$, guaranteeing reproducibility.

3.6. Graph construction and export

`nlp_graph.py` builds a `NetworkX` graph. Node attribute `names` lists the canonical name followed by all alias surface forms (semicolon-separated). Edge attribute `edge_type` is written into the GraphML output. Isolated nodes (degree 0) are removed. `nlp_visualize_web.py` generates `all_networks.html` containing all 37 interactive PyVis networks with colour-coded edges (green = *friendly*, red = *hostile*, grey = *neutral*) and per-chapter label-distribution statistics in a navigation sidebar.

4. Experimental Results

4.1. Coverage and runtime

The pipeline produces valid GraphML output for all 37 chapters (100%). Table 1 compares the approaches tried.

Table 1: *Approach comparison: runtime and retention decision.*

Approach	Runtime	Main limitation	Kept
Multi-model NER	minutes	False positives survive vote	No
Rule-based NER	≈4 s	Gazetteer-bounded recall	**Yes**
NLI mDeBERTa	6–8 h	Too slow for iteration	No
NLI CamemBERT	6–8 h	Too slow for iteration	No
LLM API	n/a	Free-tier quota	No
Sentiment XLM-R	≈8 min	Sentiment $\neq$ relation	**Yes**

4.2. End-to-end trace: chapter `paf0`

Chapter `paf0` opens *Prélude à Fondation*: Hari Seldon arrives on Trantor and is summoned before Emperor Cléon. We trace it through every pipeline stage.

The rule-based NER finds eight surface forms: *Seldon* (33), *Cléon* (18), *Demerzel* (12), *Sire* (12), *Hari Seldon* (5), *Eto Demerzel* (3), *Hummin* (2), *Alban Wellis* (1). Counts are raw mention frequencies before alias merging.

Alias merging collapses *Sire*, *Empereur*, *l'Empereur*, and *Cléon Ier* into *Cléon*; *Eto Demerzel* into *Demerzel*; and *Hari Seldon* into *Seldon* via the single-to-multi rule. Four canonical nodes remain: *Seldon* (38), *Cléon* (30), *Demerzel* (15), *Hummin* (2). *Alban Wellis* is discarded (count 1, below `min_occurrences` = 2).

At `distance_max` = 150, six pairs survive with weights: Seldon–Cléon (1199), Demerzel–Seldon (791), Demerzel–Cléon (547), Seldon–Hummin (300), Demerzel–Hummin (144), Hummin–Cléon (117). The Seldon–Cléon weight is the highest because the encyclopaedic chapter header names both characters repeatedly before the throne-room scene begins.

Table 2 shows per-snippet scores for two pairs. For Cléon–Seldon, snippet 1 covers the encyclopaedic header describing Cléon's reign and Seldon's birth year; the model scores it negative ($s = -0.171$, hostile). Snippets 2 and 3 are expository and score near zero ($s = +0.012, +0.022$, neutral). Hostile weight 0.171 beats neutral weight 0.034: the pair is labelled hostile. For Demerzel–Hummin, snippet 1 crosses the threshold at $s = +0.108$ (friendly); snippet 2 falls below ($s = -0.047$, neutral). Friendly weight 0.108 beats neutral 0.047: friendly wins.

Six edges, all retained. Five are hostile; only Demerzel–Hummin is friendly. The hostile label on Hummin–Seldon is a misclassification: Hummin is Seldon's protector here, but their snippets describe tense, conspiratorial dialogue. Sentiment on expository prose tracks surface tension, not underlying alliance.

Table 2: *Per-snippet sentiment scores for two pairs in* `paf0`. $s = p_{pos} - p_{neg}$*; label applies when* $|s| > 0.1$.

Pair	Snip.	p_{neg}	p_{neu}	p_{pos}	Vote
Cléon–Seldon	1	0.431	0.308	0.260	hostile
	2	0.382	0.224	0.394	neutral
	3	0.388	0.202	0.410	neutral
Demerzel–Hummin	1	0.208	0.476	0.316	friendly
	2	0.323	0.402	0.276	neutral

Table 3: *Graph statistics and key parameters.*

Metric	Value
Chapters processed	37 / 37 (100%)
Nodes per chapter	2–10
Edges per chapter	1–40
Anti-dictionary	1012 entries
`distance_max`	150 tokens
`MAX_CONTEXTS_PER_PAIR`	5
`sentiment threshold` $\|s\|$	0.1
`FUZZY_THRESHOLD`	88

4.3. Graph statistics

4.4. Per-chapter graph statistics

Table 4 reports node and edge counts by book. The range is wide: paf9 and lca2 each have one edge; paf15 and lca17 reach 30 and 29 respectively. Both books share a median of 5 nodes and 8–9 edges, despite PAF's political-intrigue structure and LCA's detective format. The three chapters at the high end (paf15, paf17, lca17) are all late-book ensemble scenes with four or more named characters sharing the same location.

Table 4: *Per-chapter graph size by book.*

	Nodes			Edges		
Book	**Min**	**Med**	**Max**	**Min**	**Med**	**Max**
PAF (19 ch.)	2	5	9	1	9	30
LCA (18 ch.)	2	5	9	1	8	29
Overall	2	5	9	1	8	30

4.5. Label distribution

The actual label counts across all 37 chapters are: hostile = 239, neutral = 82, friendly = 58. Table 5 breaks this down by book.

PAF averages 6.6 hostile edges per chapter against 2.1 friendly; LCA averages 6.3 hostile and 1.1 friendly. Hostile density is nearly identical across both books, but PAF produces nearly double the friendly edges per chapter, driven by the sustained Seldon/Dors alliance across multiple chapters. Neutral counts are equal at 41 per book despite PAF having one extra chapter.

Hostile interactions dominate, which is a correct finding, not a system failure. *Les Cavernes d'Acier* contributes a great number of hostile edges through its detective-antagonist structure (Baley opposing robot factions throughout most of the book, serial confrontations). *Prélude à Fondation* contributes hostile interactions through political coercion (Seldon meets the Emperor under duress) and factional conflict across Trantor's sectors. The low friendly count reflects Asimov's narrative economy: lasting alliances are few (Seldon/Dors, Baley/Daneel) and their positive valence must overcome the negative sentiment accumulated across the rest of the chapter's snippets in the weighted vote.

Table 5: *Label distribution by book.*

Book	Friendly	Hostile	Neutral
PAF (19 ch.)	39	125	41
LCA (18 ch.)	19	114	41
Total	58	239	82

4.6. Window-size ablation

`distance_max` sets the token radius within which two character mentions count as co-occurring. A wider window raises edge recall but admits ambient pairs that never truly interact. We re-ran the full pipeline at three window sizes and submitted each to the Kaggle leaderboard for the AMS competition. Table 6 reports the results.

Table 6: *Effect of* `distance_max` *on Kaggle score. All other parameters fixed; submissions made on the same day to control for leaderboard variance.*

`distance_max` (tokens)	Kaggle score
25	0.58808
75	0.57708
150	**0.59511**

All three settings fall within $\Delta \approx 0.018$, so the pipeline is not strongly sensitive to this parameter. The relationship is non-monotonic: 25 tokens beats 75, and 150 beats both. The dip at 75 is consistent with a window too wide to preserve dialogic locality but too narrow to bridge a paragraph of intervening narration. The 150-token setting, used for the final submission, gives the best score, suggesting that interactions in Asimov's prose typically span one to two paragraphs before the second character is named again. Values above 150 were not tested due to leaderboard submission limits; a sweep over 50/100/150/200/300 against a held-out gold annotation would give a tighter estimate.

4.7. Qualitative validation

The utility `print_validation_report` in `nlp_relation.py` prints the top N hostile and friendly edges with their driving snippet. This spot-check revealed consistent edge assignments: in `paf0`, Seldon meets the Emperor under clearly coercive circumstances; the sentiment model correctly assigns *hostile* to that pair. The report also showed that several *neutral* edges in PAF correspond to brief chapter appearances where two characters merely share a location, which is appropriate behaviour given the threshold.

4.8. Error Analysis

NER errors. The rule-based backend eliminates model-specific biases but introduces its own failure modes. The capitalisation heuristic catches non-names: common nouns capitalised after a colon or chapter heading (*la Crise*, *le Conseil*) can pass if not in the blocklist. Recall is bounded by gazetteer completeness: a minor character referred to only by an unfamiliar form not in the gazetteer is silently missed.

Alias fragmentation. Short-form aliases whose keyword set does not satisfy the subset rule require a manual override. For example, *Dors* and *Dors Venabili* remain separate nodes automatically because the single token *dors* is a subset of $\{$*dors, venabili*$\}$ but the fuzzy ratio `fuzz_ratio("dors", "dors venabili")` ≈ 53 falls below threshold 88. Until the override file is updated, edge weights are split across two nodes, underestimating the true interaction count of the character.

Sentiment $\neq$ relation on expository prose. The sentiment analysis model we used was trained on social-media text. Asimov's literary register (long philosophical debates, indirect irony, strategic information withholding) is out of distribution. An argument expressed through polite disagreement produces a weakly negative sentiment score, which may fall below 0.1 in absolute value and default to *neutral* rather than *hostile*. Conversely, an affectionate epistolary opening can momentarily push sentiment positive within a snippet even for a pair whose overall relationship is hostile.

Sentiment thresholds not optimised. The per-snippet decision boundary $|s| > 0.1$ was set empirically. Without a held-out gold-standard annotation, there is no principled basis for this value. A lower boundary increases *friendly* and *hostile* recall at the cost of precision; a higher one increases precision but collapses borderline snippets to *neutral* votes. The `print_validation_report` utility is the only quality signal currently available.

Co-occurrence window over-linking. A 150-token window can span paragraph boundaries. When characters A, B, and C share a scene, all three pairs receive co-occurrence edges even if only A and B interact directly. The relation typing module then classifies a pair with no real interaction; these typically yield weak per-snippet sentiment scores that vote *neutral*, but they inflate edge counts and occasionally produce spurious labels from ambient narrative tone.

5. Conclusion

We presented a character network extraction pipeline developed through two documented iterative journeys. For NER, a multi-model ensemble (spaCy + Stanza + Flair) was replaced by a rule-based backend completing all 37 chapters in $\sim$4 seconds without model downloads. For relation typing, two NLI models (6–8 hours on CPU each) and LLM APIs (free-tier quotas) were rejected before settling on a sentiment-based classifier running in $\sim$8 minutes.

The system achieves 100% corpus coverage. The hostile-dominant label distribution (239 hostile, 82 neutral, 58 friendly) is consistent with the political intrigue and detective antagonism of the two Asimov novels.

The main strengths are speed, reproducibility (deterministic cache), and zero annotation requirement. The main weaknesses are: (i) the sentiment model is out-of-distribution for literary French, causing misclassification of ironic or indirect interactions; (ii) gazetteer-bounded recall in the rule-based NER; (iii) the sentiment decision boundary was not tuned on held-out data.

Future work should collect a gold-standard annotation (even 30–50 character pairs) to enable principled threshold selection and model comparison. Adding dialogue-structure features, such as speaker identification and speech-act type, would provide relational signals complementary to surface sentiment. Extending the gazetteer to the full Foundation series would support macro-level analysis of how alliances evolve across a multi-book narrative arc.

6. References

[1] F. Moretti, *Graphs, Maps, Trees*. London: Verso, 2005.

[2] A. Bonato *et al.*, "Mining and modeling character networks," in *Proc. WAW*, LNCS 10088, Springer, 2016, pp. 100–114.

[3] D. K. Elson, N. Dames, and K. R. McKeown, "Extracting social networks from literary fiction," in *Proc. ACL*, 2010, pp. 138–147.

[4] J. Devlin *et al.*, "BERT: Pre-training of deep bidirectional transformers," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[5] W. Yin, J. Hay, and D. Roth, "Benchmarking zero-shot text classification," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3905–3914.

[6] M. Laurer *et al.*, "Less annotating, more classifying," *Political Analysis*, vol. 31, no. 1, pp. 1–13, 2022.

[7] V. Labatut and X. Bost, "Extraction and analysis of fictional character networks: A survey," *ACM Comput. Surv.*, vol. 52, no. 5, pp. 1–40, 2019.

[8] N. Dekker, B. Karsdorp, and F. Karsdorp, "Evaluating named entity recognition tools for extracting social networks from novels," in *Proc. DHd*, 2019.

[9] F. Barbieri *et al.*, "TweetEval: Unified benchmark and comparative evaluation for tweet classification," in *Proc. EMNLP Findings*, 2020, pp. 1644–1650.

[10] S. Grayson *et al.*, "The sense and sensibility of different sliding windows in constructing co-occurrence networks from literature," in *Proc. Digital Humanities*, 2016.

[11] H. Vala *et al.*, "Mr. Bennet, his coachman, and the Archbishop walk into a bar but only one of them gets recognized: On the difficulty of detecting characters in literary texts," in *Proc. EMNLP*, 2015, pp. 769–778.

[12] E. T. Nalisnick and H. S. Baird, "Character-to-character sentiment analysis in Shakespeare's plays," in *Proc. ACL*, 2013, pp. 479–483.

[13] J. Wei *et al.*, "Revisiting large language models as zero-shot relation extractors," in *Proc. EMNLP Findings*, 2023, pp. 6877–6892.

[14] L. Martin *et al.*, "CamemBERT: A tasty French language model," in *Proc. ACL*, 2020, pp. 7203–7219.

[15] F. Béchet *et al.*, "BookNLP-fr: French part of the Multilingual BookNLP project," Observatoire des humanités numériques, ENS Paris, 2024.

[16] A. Landragin *et al.*, "The elephant in the coreference room: Resolving coreference in full-length French fiction works," in *Proc. CRAC Workshop, ACL*, 2025.